

Report: Vessel Speed Optimization Under Oil Price Volatility

Motivation

Container shipping firms face a fundamental tradeoff between fuel cost and delivery speed: sailing faster reduces transit time but sharply increases fuel consumption, while sailing slower lowers bunker use but raises the cost of inventory tied up in transit. On services with fixed weekly departures, excessive slow steaming may require an additional vessel to preserve schedule frequency, meaning the speed that appears cheapest on fuel alone may no longer be optimal once fleet requirements are considered. This is the central operational problem examined in this report.

The problem is especially relevant given current geopolitical exposure. The Strait of Hormuz is one of the world's most critical oil chokepoints, and in its April 2026 outlook the EIA projected Brent crude rising from \$81 per barrel in Q1 2026 to a peak of \$115 in Q2, reflecting conflict-driven disruption risk (EIA, 2026a; 2026b). In this environment, vessel speed becomes a strategic operating decision rather than a routine scheduling choice, and recent oil market forecasts reinforce the practical importance of analyzing it under fuel price volatility.

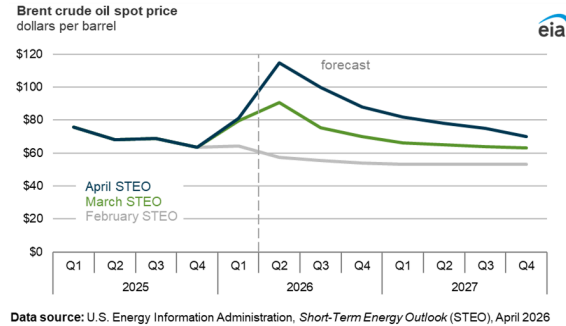


Figure 1. Brent crude oil spot price forecast.

Background

Slow steaming reduces fuel expenditure because consumption rises approximately with the cube of vessel speed, meaning even moderate reductions generate substantial savings (IMO, 2023). However, speed decisions cannot be evaluated through fuel savings alone. Container lines operate scheduled services, and slowing down too much can push round-trip time beyond a weekly threshold, requiring an additional vessel and creating a step-cost structure in which the minimum-cost speed is not necessarily the slowest feasible one. In a period marked by oil price volatility and geopolitical risk, this makes vessel speed optimization both analytically meaningful and operationally relevant.

Research question: *What is the cost-minimizing vessel speed as oil prices change, and how does the fleet-frequency constraint shift that optimum?*

Optimization Model

Layer 1 Base Speed Optimization

Decision variable: v = vessel speed (knots), subject to $12 \leq v \leq 25$

Parameters

D	<i>One – way route distance (Shanghai to LA)</i>
$v_0 = 24 \text{ kn}$	<i>Design (reference) speed</i>
p	<i>Fuel price (\$/ MT)</i>
α	<i>Fuel consumption at v_0 (MT/ day)</i>
h	<i>Daily pipeline inventory holding cost (\$/ day)</i>

Fuel consumption per day (cubic admiralty law):

$$FC(v) = \alpha \cdot \left(\frac{v}{v_0}\right)^3 [\text{MT/ day}]$$

Transit time (one way; 24 converts knots to nmi/ day):

$$T(v) = \frac{D}{24v} [\text{days}]$$

Cost components per voyage:

$$C_{fuel}(v) = p \cdot FC(v) \cdot T(v) = \frac{p\alpha D}{24v_0^3} v^2$$

$$C_{inv}(v) = h \cdot T(v) = \frac{hD}{24v}$$

Note: $C_{fuel} \propto v^2$ because $FC \propto v^3$ and $T \propto v^{-1}$, so fuel per voyage $\propto v^3 \cdot v^{-1} = v^2$. The opposing v^2 and v^{-1} terms guarantee an interior optimum.

Total cost per voyage:

$$TC(v) = \frac{p\alpha D}{24v_0^3} v^2 + \frac{hD}{24} v^{-1} = A v^2 + B v^{-1}$$

Closed-form optimal speed (set $\frac{dTC}{dv} = 0$):

$$v^* = \left[\frac{h v_0^3}{2p\alpha} \right]^{1/3}$$

Key result: $v^* \propto p^{-1/3}$. If bunker price doubles, optimal speed falls by $2^{1/3} \sim 21\%$.

Layer 2 Fleet frequency constraint

To maintain weekly departures with ~ 2 days port time each end, the number of vessels required is:

$$T_{round}(v) = \frac{2D}{24v} + 4 \text{ [days]}$$

$$N(v) = \left\lceil \frac{T_{round}(v)}{7} \right\rceil \text{ [vessels]}$$

Each vessel carries weekly charter cost c_{cap} (\$/ vessel/ week). c_{op} is the operating cost to maintain the fleet including crew.

Extended weekly total cost:

$$TC_{ext}(v) = 2(A v^2 + B v^{-1}) + N(v) \cdot (c_{cap} + 7c_{op})$$

$N(v)$ is a ceiling step function: it jumps by 1 each time $T_{round}(v)$ crosses a multiple of 7 days (e.g. reducing speed from 18 to 17 kn may push round-trip from 41 to 43 days, adding a 7th vessel). The smooth Layer 1 optimum v^* may no longer minimize TC_{ext} .

Connection to course concepts

The core modeling structure is a direct application of the inventory-theoretic tradeoff framework covered in class. In the standard EOQ model, total cost is minimized by balancing opposing costs in the decision variable, yielding a closed-form power-of-a-ratio expression from a first-order condition. Our model has exactly the same structure: fuel cost and pipeline inventory holding cost pull in opposite directions as speed changes, and the optimal speed formula emerges from identical logic. The exponent differs from EOQ's square root, reflecting the cubic fuel-speed relationship, but the economic intuition is the same. The pipeline inventory interpretation also connects directly to course material on lead times — goods at sea are inventory, transit time is the replenishment lead time, and the holding cost term is precisely the pipeline stock cost from multi-stage inventory models. Layer 2 introduces a fixed cost structure mirroring two further course themes: the fixed ordering cost in EOQ, where each additional vessel plays the role of a fixed charge, and the fixed-charge network design problem, where each vessel functions as a facility whose activation introduces non-convexities analogous to those in facility location models.

Layer 3 Scenario Analysis

Both layers are evaluated across five bunker price scenarios (\$1000, \$1500, \$2000, \$2500, \$3000 per MT), spanning the historical range plus stress case. Output scenario: v^* , N^* , full cost breakdown, and % saving vs. design speed.

Key Parameters & Data Sources

All parameters were sourced from published industry databases, academic literature, and manufacturer technical documentation. Where sources reported a range, the midpoint was adopted as the base-case estimate to minimize directional bias, with bounds revisited in sensitivity analysis.

Route distance ($D = 5,500$ nmi) reflects the standard Shanghai–Los Angeles trans-Pacific routing from SeaRates.com. Design speed ($v_0 = 24$ knots) is taken from Maersk fleet specifications and serves as the anchor for all fuel consumption scaling. Fuel consumption at design speed ($\alpha = 165$ MT/day) is the midpoint of the 150–180 MT/day range from Wärtsilä engine performance curves for two-stroke diesels typical of 12,000–15,000 TEU vessels. Bunker price scenarios span \$1,000–\$3,000/MT, drawing on Ship & Bunker and EIA data, with \$2,000/MT as the base case and \$3,000/MT representing a Hormuz disruption stress scenario.

Daily holding cost ($h = \$168,000/\text{day}$) was computed as $h = i \times (\text{cargo value per TEU} \times \text{TEUs loaded}) / 365$, applying a 20% annual holding rate (Journal of Business Logistics), \$30,000 per TEU cargo value (US Census Bureau), and 10,200 loaded TEUs at an 85% load factor. Daily operating cost ($c_{op} = \$20,000/\text{day}$) is the midpoint of Drewry's \$15,000–\$25,000 range and is treated as a fixed fleet cost in Layer 2. Charter cost ($c_{cap} = \$455,000/\text{week}$) is derived from Clarkson's midpoint time-charter rate of \$65,000/day, active only when slow steaming triggers a fleet-size increase. Port time is fixed at two days per call based on MarineTraffic AIS data, giving four days total per round trip.

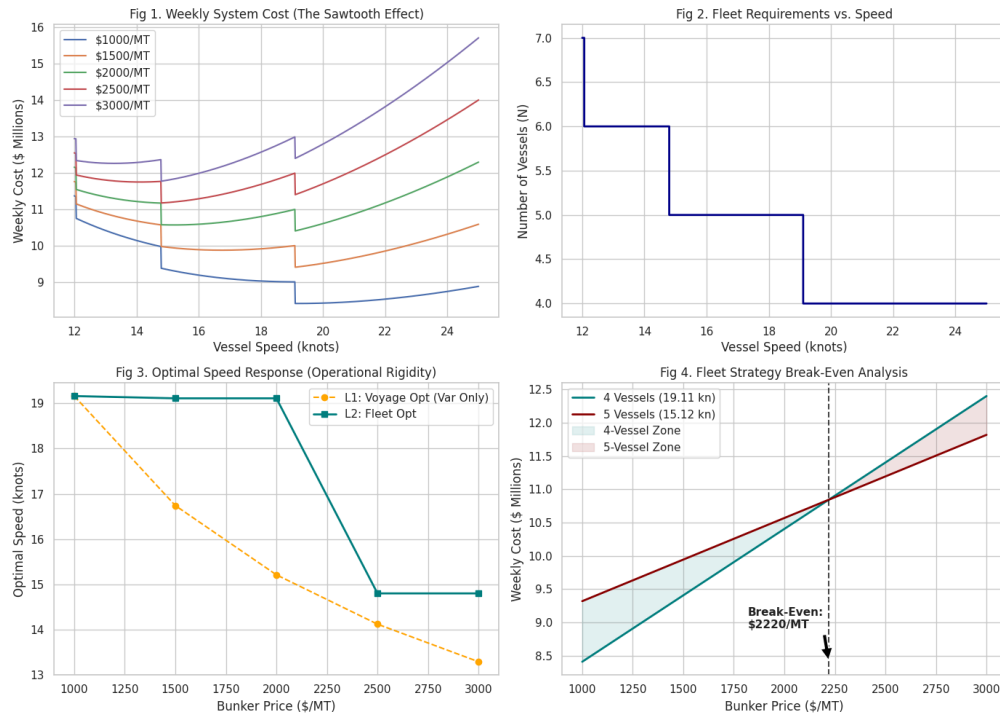
<i>Parameter</i>	<i>Symbol</i>	<i>Value</i>
<i>Route distance</i>	D	<i>5,500 nmi</i>
<i>Design speed</i>	v_0	<i>24 kn</i>
<i>Fuel at design speed</i>	α	<i>165 MT/day</i>
<i>Bunker price</i>	p	<i>\$1000–\$3,000/MT</i>
<i>Holding cost rate</i>	i	<i>20%/yr</i>
<i>Cargo value per TEU</i>	—	<i>\$30,000</i>
<i>Loaded TEUs</i>	—	<i>10,200</i>
<i>Daily holding cost</i>	h	<i>\$168,000/day</i>
<i>Daily OPEX</i>	c_{op}	<i>\$20,000/day</i>
<i>Charter cost</i>	c_{cap}	<i>\$455,000/week</i>
<i>Port time</i>	—	<i>2 days/port</i>

Simulation and Inference

To evaluate the strategic tradeoff between fuel consumption and fleet deployment, the parameters were simulated across the two modeling layers. Layer 1 evaluated the continuous, theoretical voyage optimization (variable costs only), while Layer 2 applied the discrete fleet-frequency constraint (system-wide variable and fixed costs). The core results of the simulation are summarized in Table 2.

Optimization Results by Bunker Price Scenario

Bunker Price (\$/MT)	L1 Opt Speed (kn)	L2 Opt Speed (kn)	Fleet Size (N)	Weekly Cost (\$M)	Transit Time (Days)
1000	19.16	19.16	4	8.407	12.0
1500	16.74	19.11	4	9.406	12.0
2000	15.21	19.11	4	10.404	12.0
2500	14.12	14.80	5	11.173	15.5
3000	13.29	14.80	5	11.772	15.5



The simulation reveals that vessel speed optimization under scheduled service constraints is not a smooth, continuous mathematical function. Instead, total weekly system costs exhibit a distinct "sawtooth" pattern. Gradual speed reductions yield steady fuel savings until the transit time pushes the round-trip duration past a multiple of seven days (e.g., 28 days for a 4-vessel loop).

Once this threshold is crossed, the schedule breaks. To maintain a weekly port call frequency at slower speeds, the shipping line is forced to charter a 5th vessel. This triggers a severe step-cost penalty of \$595,000 per week (\$455,000 in charter fees plus \$140,000 in fleet-wide OPEX). Consequently, the absolute minimum weekly cost is rarely the slowest feasible speed, but rather the speed situated just above a fleet-size threshold.

The data highlights a significant divergence between theoretical voyage optimization and practical fleet deployment, resulting in severe operational rigidity.

Under the Layer 1 unconstrained model, optimal speed decreases continuously as bunker prices rise; for example, moving from \$1,000/MT to \$2,000/MT drops the theoretical optimal speed from 19.16 knots to 15.21 knots. However, in the Layer 2 operational model, the speed remains completely static at exactly 19.11 knots.

This 19.11-knot floor is the mathematical minimum speed required to execute a 5,500 nmi round-trip, including 4 days of port time, within 28 days using exactly 4 vessels. At \$1,500/MT and \$2,000/MT, the fuel savings achieved by slow steaming are entirely eclipsed by the massive capital step-cost of deploying a 5th vessel. Management is economically incentivized to absorb significantly higher fuel bills rather than break the schedule and incur the capital expense of an extra ship.

A break-even analysis was conducted to map the minimum cost of a 4-vessel strategy (anchored at 19.11 knots) against a slower 5-vessel strategy (anchored at 14.80 knots). The simulation identifies this strategic tipping point at \$2,219.61/MT.

Below this extreme threshold, the firm should prioritize capital efficiency, maintaining a 4-vessel fleet and a 12-day transit time. Only when bunker prices exceed \$2,220/MT does the cost of fuel become so overwhelmingly punitive that the firm optimally transitions to a 5-vessel deployment, accepting a 15.5-day transit time.

The primary inference drawn from this model is that long-haul container networks with strict frequency constraints are highly resistant to slow-steaming policies unless forced by catastrophic market shocks. Because current and historical stress-case bunker prices rarely approach the \$2,220/MT threshold, liner operators on the Asia–US corridor must prioritize capital efficiency and schedule preservation over fuel efficiency. The threshold to justify adding a vessel solely for the purpose of fuel conservation requires an unprecedented anomaly in global energy markets.

In practice, liner operators do not always sail at one fixed “optimal” speed. Vessel speed is adjusted dynamically in response to changing operating conditions. For example, ships may slow down or reroute in adverse weather for safety reasons, but operators may later increase speed to recover lost time and maintain schedule reliability (Brouer et al., 2013; IMO, 2007). Similarly, ships may reduce speed when port congestion or berth unavailability makes early arrival inefficient, since just-in-time arrival can reduce waiting time and unnecessary fuel burn (Global Industry Alliance to Support Low Carbon Shipping, 2021). For this reason, the speed derived in this model should be interpreted as a planning benchmark under normal conditions rather than as a rule that carriers follow exactly at all times.

Limitations

Several simplifying assumptions limit the generalizability of this model. First, the cubic admiralty law provides a useful approximation of the fuel-speed relationship but does not account for vessel-specific engine efficiency curves, hull fouling, or sea state conditions, all of which can cause real-world fuel consumption to deviate meaningfully from the theoretical benchmark. Second, the model treats cargo value, load factor, and holding cost rate as fixed constants, whereas in practice these vary by commodity type, season, and shipper. Third, the fleet-frequency constraint assumes a rigid weekly schedule with uniform port times, abstracting away the possibility of port congestion, berth delays, or schedule flexibility that liner operators routinely manage. Fourth, the charter cost used to price the step-cost penalty reflects a normalized mid-cycle rate and may understate costs during supply-constrained markets. Finally, the model is single-route and single-commodity, and does not capture network interdependencies such as vessel repositioning, multi-port rotations, or backhaul economics that significantly influence real fleet deployment decisions.

Conclusion

This project demonstrates that vessel speed optimization on long-haul liner routes is a fundamentally non-smooth problem once fleet-frequency constraints are introduced. While Layer 1 produces an intuitive result — optimal speed falls predictably as bunker prices rise — Layer 2 reveals this theoretical optimum is largely inaccessible in practice. The discrete step-cost of deploying an additional vessel pins operators to the minimum speed compatible with their current fleet size, regardless of moderate fuel price movements. The break-even analysis quantifies this rigidity precisely: only when bunker prices exceed approximately \$2,220/MT does a fifth vessel become justifiable on cost grounds alone. Since this threshold sits well above historical norms and most stress-case scenarios, capital efficiency and schedule preservation should dominate speed decisions under all but the most extreme energy market disruptions. More broadly, this analysis illustrates how operational constraints can render theoretically optimal solutions practically irrelevant, and underscores the value of layered modeling when advising on strategic decisions in capital-intensive industries.

References

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Appendix

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Set visualization style

sns.set_theme(style="whitegrid")

plt.rcParams.update({'figure.figsize': (14, 10), 'font.size': 11})

# =====

# 1. PARAMETERS

# =====

PARAMS = {

    'D': 5500,          # Route distance (nmi)

    'v0': 24,          # Design speed (knots)

    'alpha': 165,      # Fuel consumption at design speed (MT/day)

    'h': 168000,       # Daily inventory holding cost ($/day)

    'cop': 20000,      # Daily vessel operating cost ($/day)

    'ccap_week': 455000, # Weekly charter cost ($/vessel)

    'port_time': 4     # Total port time per round trip (days)

}
```



```

# Updated Price Brackets

PRICES = [1000, 1500, 2000, 2500, 3000]

SPEED_RANGE = np.linspace(12, 25, 1000)

# =====

# 2. CORE FUNCTIONS

# =====

def calculate_costs(v, p, params):

    """Calculates Layer 1 (Voyage) and Layer 2 (System) costs based on the
    corrected accounting model."""

    # Layer 1: Voyage Variable Costs (1 One-Way Trip)

    transit_time = params['D'] / (24 * v)

    c_fuel = p * params['alpha'] * (v / params['v0'])**3 * transit_time

    c_inv = params['h'] * transit_time

    # OPEX is removed from here; it is a fixed fleet cost.

    tc_voyage_var = c_fuel + c_inv

    # Layer 2: Fleet Frequency Metrics

    t_round = (2 * params['D'] / (24 * v)) + params['port_time']

    n_vessels = np.ceil(t_round / 7)

    # Weekly System Cost

```

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    weekly_variable_costs = 2 * tc_voyage_var # 2 one-way trips per week
    guaranteed

    weekly_fixed_costs = n_vessels * (params['ccap_week'] + (7 *
params['cop'])) # Applied 24/7 across the fleet

    tc_ext = weekly_variable_costs + weekly_fixed_costs

    return tc_voyage_var, tc_ext, n_vessels

def run_scenario_simulation():

    """Runs the simulation across all defined prices and speeds."""

    results = []

    detailed_curves = {}

    for p in PRICES:

        l1_costs, l2_costs, n_counts = [], [], []

        for v in SPEED_RANGE:

            tc_v, tc_e, n = calculate_costs(v, p, PARAMS)

            l1_costs.append(tc_v)

            l2_costs.append(tc_e)

            n_counts.append(n)

        idx_l1, idx_l2 = np.argmin(l1_costs), np.argmin(l2_costs)

```

```

v_star_l1, v_star_l2 = SPEED_RANGE[idx_l1], SPEED_RANGE[idx_l2]

results.append({

    "Bunker Price ($/MT)": p,

    "L1 Opt Speed (kn)": round(v_star_l1, 2),

    "L2 Opt Speed (kn)": round(v_star_l2, 2),

    "Fleet Size (N)": int(n_counts[idx_l2]),

    "Weekly Cost ($M)": round(l2_costs[idx_l2] / 1e6, 3),

    "Transit Time (Days)": round(PARAMS['D'] / (24 * v_star_l2),
1)

    })

    detailed_curves[p] = {'speed': SPEED_RANGE, 'l2_cost': l2_costs,
'n': n_counts}

return pd.DataFrame(results), detailed_curves

def find_break_even():

    """Finds the price where the 5-vessel strategy becomes cheaper than 4
vessels."""

    p_range = np.linspace(1000, 3000, 2000)

    costs_4, costs_5 = [], []

    break_p = None

    for p in p_range:

```

```

        tc_4 = calculate_costs(19.11, p, PARAMS)[1] # Min speed for 4
vessels

        tc_5 = calculate_costs(15.12, p, PARAMS)[1] # Optimal lower speed
for 5

        costs_4.append(tc_4 / 1e6)

        costs_5.append(tc_5 / 1e6)

    if tc_5 < tc_4 and break_p is None:

        break_p = p

    return break_p, p_range, np.array(costs_4), np.array(costs_5)

# =====

# 3. VISUALIZATION MODULE

# =====

def plot_results(detailed_curves, df_results, break_even_data):

    fig, axs = plt.subplots(2, 2)

    # Plot 1: Sawtooth Curves

    for p in PRICES:

        axs[0, 0].plot(detailed_curves[p]['speed'], [c/1e6 for c in
detailed_curves[p]['l2_cost']], label=f'${p}/MT')

    axs[0, 0].set_title("Fig 1. Weekly System Cost (The Sawtooth Effect)")

    axs[0, 0].set_xlabel("Vessel Speed (knots)")

```

```

    axs[0, 0].set_ylabel("Weekly Cost ($ Millions)")

    axs[0, 0].legend()

    # Plot 2: Fleet Step Function (Updated to use a valid price key, 2000)

    base_price_for_plot = 2000

    axs[0, 1].step(detailed_curves[base_price_for_plot]['speed'],
detailed_curves[base_price_for_plot]['n'], where='post', color='darkblue',
lw=2)

    axs[0, 1].set_title("Fig 2. Fleet Requirements vs. Speed")

    axs[0, 1].set_xlabel("Vessel Speed (knots)")

    axs[0, 1].set_ylabel("Number of Vessels (N)")

    # Plot 3: Sticky Speed

    axs[1, 0].plot(df_results["Bunker Price ($/MT)"], df_results["L1 Opt
Speed (kn)"], 'o--', label='L1: Voyage Opt (Var Only)', color='orange')

    axs[1, 0].plot(df_results["Bunker Price ($/MT)"], df_results["L2 Opt
Speed (kn)"], 's-', label='L2: Fleet Opt', color='teal', lw=2)

    axs[1, 0].set_title("Fig 3. Optimal Speed Response (Operational
Rigidity)")

    axs[1, 0].set_xlabel("Bunker Price ($/MT)")

    axs[1, 0].set_ylabel("Optimal Speed (knots)")

    axs[1, 0].legend()

    # Plot 4: Break-Even Analysis

    break_p, p_range, costs_4, costs_5 = break_even_data

```

```

    axs[1, 1].plot(p_range, costs_4, label='4 Vessels (19.11 kn)',
color='teal', lw=2)

    axs[1, 1].plot(p_range, costs_5, label='5 Vessels (15.12 kn)',
color='darkred', lw=2)

    if break_p:

        axs[1, 1].axvline(x=break_p, color='black', linestyle='--',
lw=1.5, alpha=0.7)

        axs[1, 1].fill_between(p_range, costs_4, costs_5, where=(costs_4 <
costs_5), color='teal', alpha=0.1, label='4-Vessel Zone')

        axs[1, 1].fill_between(p_range, costs_4, costs_5, where=(costs_5
<= costs_4), color='darkred', alpha=0.1, label='5-Vessel Zone')

        # Adjusted label position slightly so it looks clean with the new
higher break-even price

        axs[1, 1].annotate(f'Break-Even:\n${break_p:.0f}/MT', xy=(break_p,
min(costs_4)), xytext=(break_p - 400, min(costs_4) + 0.5),

                        arrowprops=dict(facecolor='black',
shrink=0.05), fontweight='bold')

    axs[1, 1].set_title("Fig 4. Fleet Strategy Break-Even Analysis")

    axs[1, 1].set_xlabel("Bunker Price ($/MT)")

    axs[1, 1].set_ylabel("Weekly Cost ($ Millions)")

    axs[1, 1].legend()

plt.tight_layout()

plt.show()

```

```

# =====

# 4. EXECUTION

# =====

if __name__ == "__main__":

    df_results, detailed_curves = run_scenario_simulation()

    break_even_data = find_break_even()

    print("--- SIMULATION RESULTS ---")

    print(df_results.to_string(index=False))

    if break_even_data[0]:

        print(f"\nBreak-Even Tipping Point: ${break_even_data[0]:.2f} /
MT")

    else:

        print("\nBreak-Even Tipping Point exceeds tested range.")

    plot_results(detailed_curves, df_results, break_even_data)

```